Analytical Chemistry Cumulative Exam

September 26, 2025

Topic: Statistics and Error Propagation

You must show your working; no credit will be given for answers without working. Partial credit will be given for each step of the solution – A correctly set up problem with minor calculator errors/miscopying numbers can still pass this cume!

Statistical tables and formulae are available at the end of the exam.

You should use the appropriate number of significant figures in all your answers.

Question 1 –Choose the appropriate statistical test

For each of the scenarios listed below, **provide the equation for the appropriate statistical test**. While example data are given, for this question **you do not need to perform any calculations** or apply the statistical test.

a) You took 7 replicate IR spectroscopic measurements of a sample of wheat to assess its protein and moisture content. One of the measurements of the moisture % appears to be higher than the others, and you are concerned that it is an outlier.

7.8, 10.2, 7.7, 8.2, 9.0, 12.7, 7.3 moisture (%w/w)

*What equation would you use to test whether you have an **outlier** measurement*?

b) A student in the lab claims that their benchtop spectrometer has a problem and is reporting the concentrations too low. You carefully prepare a 10.00 mM standard solution and measure it on their spectrometer. You used accurate values for molar absorptivity and the pathlength to convert the absorption measurements into concentrations. The results of replicate measurements are: 9.96, 10.24, 10.08, 9.96, 10.09, 9.81, 10.09, 9.97, 9.67, 9.68 mM

What equation would you use to assess the student's claim?

c) You designed a new membrane for an ion-sensitive electrode for measuring NO_3^- concentrations in natural water sources (lakes, rivers, streams). You want to know if your method is accurate, and so you compare it with a validated method from the Arkansas Water Resources Center's water sample analysis lab.

	Ion-sensitive electrode (NO_3^- mg/L)	Validated method (NO_3^- mg/L)
Sample 1	0.25	0.63
Sample 2	41.2	45.0
Sample 3	…	…
…		

*What equation (test) should you use to test the **accuracy** of your method vs the validated method?*

d) Your lab bought a new pre-amplifier for your potentiostat, which you have been told gives more precise current readings for low currents. You are excited, but cautious, and want to test this claim. You make up a low concentration solution and measure it in replicate. You compare the data with measurements you previously took for a similar solution and electrode. Note: as you are measuring different solutions, their concentrations may not be identical.

Peak current without pre-amplifier (nA)	Peak current with pre-amplifier (nA)
14.2	14.4
11.5	13.1
15.6	13.0
14.1	14.4
13.5	14.3

What is the equation of the statistical test you should perform to determine if precision is improved?

Question 2. Error Propagation

You are making a solution of NaOH (39.997 g/mol) in a vial and want to know the concentration, the pOH AND the uncertainty in their values. You first weigh the empty vial as 23.06 ±0.05 g, next you add your NaOH to the vial and the total weight of the vial is now 23.61 ±0.05 g. Finally, you use a volumetric pipette to add 20.0±0.3 mL of water to the vial.

Determine the **NaOH concentration, pOH** in the solution and **the absolute error** in these quantities. Give your answers to appropriate significant figures. You may assume the MW is a precise value with no error and that all species have activity 1. ***You must show your working***.

Expected answer format: *Working…cNaOH= 1.23±45 M, pOH= 6.7±8.9*

Question 3. Application of a statistical test

Replicate measurements of the sodium content in one batch of tomato soup give 1.90, 1.83, 1.91 & 1.94 mg/mL, while in a second batch it is 1.96, 2.01, 1.98 & 1.95 mg/mL. Is there a significant difference in the sodium level in the two batches at the 95% confidence level?

Your answer should indicate 1) whether you believe there is a significant difference, 2) the value of your calculated test statistic, and 3) the critical value that you are comparing to.

You may assume that these measurements have already been checked for outliers. Please provide your full working.

Constants, Equations, and Tables

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

$$E = x_i - x_t$$

$$E_r = \frac{x_i - x_t}{x_t} \times 100\%$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

$$s = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \bar{x})^2}{N-1}} = \sqrt{\frac{\sum_{i=1}^{N} x_i^2 - \frac{\left(\sum_{i=1}^{N} x_i\right)^2}{N}}{N-1}}$$

$$s_m = \frac{s}{\sqrt{N}}$$

$$s_{pooled} = \sqrt{\frac{\sum_{i=1}^{N_1} (x_i - \overline{x_1})^2 + \sum_{j=1}^{N_2} (x_j - \overline{x_2})^2 + \sum_{k=1}^{N_3} (x_k - \overline{x_3})^2 + \ldots}{N_1 + N_2 + N_3 + \ldots - N_t}}$$

$$s_{pooled} = \sqrt{\frac{(N_1 - 1)\, s_1^2 + (N_2 - 1)\, s_2^2 + (N_3 - 1)\, s_3^2 + \ldots}{N_1 + N_2 + N_3 + \ldots - N_t}}$$

$$y = \frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}} = \frac{e^{-z^2/2}}{\sigma\sqrt{2\pi}}$$

$$z = \frac{(x - \mu)}{\sigma}$$

$$\mu = \bar{x} \pm \frac{z\,\sigma}{\sqrt{N}}$$

$$\mu = \bar{x} \pm \frac{t\,s}{\sqrt{N}}$$

$$z = \left| \frac{\bar{x} - \mu_o}{\sigma / \sqrt{N}} \right|$$

$$t = \left| \frac{\bar{x} - \mu_o}{s / \sqrt{N}} \right|$$

$$t = \left| \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled} \sqrt{\frac{N_1 + N_2}{N_1 N_2}}} \right|$$

$$z = \left| \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{N_1 + N_2}{N_1 N_2}}} \right|$$

$$s_d = \sqrt{\frac{\sum \left(d_i - \bar{d}\right)^2}{N - 1}}$$

$$t = \frac{|\bar{d}|}{s_d / \sqrt{N}}$$

$$F = \frac{s_1^2}{s_2^2}$$

$$G_{calc} = \frac{|\text{questionable value} - \bar{x}|}{s}$$

$$RSD = \frac{s}{\bar{x}}$$

$$CV = \frac{s}{\bar{x}} \times 100\%$$

$$w = \left(x_{largest} - x_{smallest}\right)$$

	Function	Uncertainty
+/-	$y = x_1(\pm e_{x_1}) + x_2(\pm e_{x_2}) - x_3(\pm e_{x_3})$	$e_y = \sqrt{e_{x_1}^2 + e_{x_2}^2 + e_{x_3}^2}$
×/÷	$y = \dfrac{x_1(\pm e_{x_1}) \times x_2(\pm e_{x_2})}{x_3(\pm e_{x_3})}$	$\dfrac{e_y}{y} = \sqrt{\left(\dfrac{e_{x_1}}{x_1}\right)^2 + \left(\dfrac{e_{x_2}}{x_2}\right)^2 + \left(\dfrac{e_{x_3}}{x_3}\right)^2}$
	where $\%e_{x_1} = 100\% \dfrac{e_{x_1}}{x_1}, \%e_{x_2} = 100\% \dfrac{e_{x_2}}{x_2}$, $\%e_{x_3} = 100\% \dfrac{e_{x_3}}{x_3}, \%e_y = 100\% \dfrac{e_y}{y}$	or $\%e_y = \sqrt{(\%e_{x_1})^2 + (\%e_{x_2})^2 + (\%e_{x_3})^2}$
exponentiation	$y = x(\pm e_x)^a$	$\dfrac{e_y}{y} = a\,\dfrac{e_x}{x}$
		or $\%e_y = a\ \%e_x$ where $\%e_y = 100\% \dfrac{e_y}{y}, \%e_x = 100\% \dfrac{e_x}{x}$
logarithm	$y = \log_{10}(x(\pm e_x))$	$e_y = \dfrac{1}{\ln 10}\dfrac{e_x}{x} = 0.43429\ \dfrac{e_x}{x}$
	$y = \ln(x(\pm e_x))$	$e_y = \dfrac{e_x}{x}$
antilogarithm	$y = \text{antilog}_{10}(x(\pm e_x)) = 10^{x(\pm e_x)}$	$\dfrac{e_y}{y} = (\ln 10)\, e_x \approx 2.3026\, e_x$

TABLE 4-5 Critical values of *G* for rejection of outlier

Number of observations	*G* (95% confidence)
4	1.463
5	1.672
6	1.822
7	1.938
8	2.032
9	2.110
10	2.176
11	2.234
12	2.285
15	2.409
20	2.557

$G_{calculated} = |questionable\ value - mean|/s$. *If* $G_{calculated} > G_{table}$, *the value in question can be rejected with 95% confidence. Values in this table are for a one-tailed test, as recommended by ASTM.*

TABLE 4-4 Values of Student's *t*

Degrees of freedom	Confidence level (%) 50	90	95	98	99	99.5	99.9
1	1.000	6.314	12.706	31.821	63.656	127.321	636.578
2	0.816	2.920	4.303	6.965	9.925	14.089	31.598
3	0.765	2.353	3.182	4.541	5.841	7.453	12.924
4	0.741	2.132	2.776	3.747	4.604	5.598	8.610
5	0.727	2.015	2.571	3.365	4.032	4.773	6.869
6	0.718	1.943	2.447	3.143	3.707	4.317	5.959
7	0.711	1.895	2.365	2.998	3.500	4.029	5.408
8	0.706	1.860	2.306	2.896	3.355	3.832	5.041
9	0.703	1.833	2.262	2.821	3.250	3.690	4.781
10	0.700	1.812	2.228	2.764	3.169	3.581	4.587
15	0.691	1.753	2.131	2.602	2.947	3.252	4.073
20	0.687	1.725	2.086	2.528	2.845	3.153	3.850
25	0.684	1.708	2.060	2.485	2.787	3.078	3.725
30	0.683	1.697	2.042	2.457	2.750	3.030	3.646
40	0.681	1.684	2.021	2.423	2.704	2.971	3.551
60	0.679	1.671	2.000	2.390	2.660	2.915	3.460
120	0.677	1.658	1.980	2.358	2.617	2.860	3.373
∞	0.674	1.645	1.960	2.326	2.576	2.807	3.291

In calculating confidence intervals, σ may be substituted for s in Equation 4-7 if you have a great deal of experience with a particular method and have therefore determined its "true" population standard deviation. If σ is used instead of s, the value of t to use in Equation 4-7 comes from the bottom row of this table.

Values of t in this table apply to two-tailed tests illustrated in Figure 4-9a. The 95% confidence level specifies the regions containing 2.5% of the area in each wing of the curve. For a one-tailed test, we use values of t listed for 90% confidence. Each wing outside of t for 90% confidence contains 5% of the area of the curve.

TABLE 4-4 Critical values of $F = s_1^2/s_2^2$ at 95% confidence level

Degrees of freedom for s_2	Degrees of freedom for s_1: 2	3	4	5	6	7	8	9	10	12	15	20	30	∞
2	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5
3	9.55	9.28	9.12	9.01	8.94	8.89	8.84	8.81	8.79	8.74	8.70	8.66	8.62	8.53
4	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.75	5.63
5	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.50	4.36
6	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.81	3.67
7	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.58	3.51	3.44	3.38	3.23
8	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.08	2.93
9	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.86	2.71
10	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.84	2.77	2.70	2.54
11	3.98	3.59	3.36	3.20	3.10	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.57	2.40
12	3.88	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.47	2.30
13	3.81	3.41	3.18	3.02	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.38	2.21
14	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.31	2.13
15	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.25	2.07
16	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.19	2.01
17	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.15	1.96
18	3.56	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.11	1.92
19	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.07	1.88
20	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.04	1.84
30	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.84	1.62
∞	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.46	1.00

Critical values of F for a one-tailed test of the hypothesis that $s_1 > s_2$. There is a 5% probability of observing F above the tabulated value.

You can compute F for a chosen level of confidence with the Excel function FINV(probability,deg_freedom1,deg_freedom2). The statement "=FINV(0.05,7,6)" reproduces the value F = 4.21 in this table. The statement "=FINV(0.1,7,6)" gives F = 3.01 for 90% confidence.

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